

Critical Rydberg electrometry with a light–matter phase transition

what Ding *et al.* and Wang *et al.* do, with the transition of Zhang *et al.* in place of the
interaction-driven one

Feasibility study and literature assessment

August 24, 2026

Abstract

Ding *et al.* [Nat. Phys. **18**, 1447 (2022)] and Wang *et al.* [arXiv:2502.19761] measure microwave electric fields by biasing a Rydberg ensemble at a critical point and reading the steep response. Their critical point is that of a *non-equilibrium* transition — the optical bistability of a driven, interacting Rydberg gas. Its order parameter is the Rydberg population; the light is a probe.

This note keeps their sensing protocol and swaps the phase transition. We use a *light–matter* transition, in which the cavity field itself is the order parameter, because the microwave field then sets the dressed-polariton splitting $\tilde{\omega}_0 = \delta + \Omega_\mu/2$ and therefore *moves the phase boundary*. Two flavours are available. **Route I** is the rotating-wave transition of Zhang *et al.* [PRL **110**, 090402 (2013)]; in a driven cavity it is realised as a superradiant *lasing* threshold, and it needs no new light–matter physics beyond a repump. **Route II** is the $\mathbb{Z}_2$ Dicke transition of Dimer *et al.* / Baden *et al.*, which requires engineered counter-rotating terms and is the natural second step.

Worked out here, with numerics. Route I has a threshold $G_{\text{th}}^2 = \kappa\Gamma[1 + D^2/(\kappa + \Gamma)^2]$ (verified against linear stability to 8 digits), and in the bad-cavity limit the emitted frequency locks to the atoms: $\partial\nu/\partial\tilde{\omega}_0 = 0.990$ while $\partial\nu/\partial\omega_{\text{cav}} = 9.9 \times 10^{-3}$, so **cavity drift is rejected** 101× at $\kappa/\Gamma = 100$. That directly answers the standing objection to critical sensing — that criticality amplifies drift as readily as signal. Reading the measurand as an optical *frequency* rather than an amplitude gives $\sqrt{S_E} = 0.070 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$ at $N = 10^6$, a factor 5.1 better than Route II at equal atom number because the lasing route uses all N atoms whereas Route II is capped by $n_b \ll N$. Route II in exchange offers a genuine ground-state critical point, ε^{-1} transduction gain and 10 dB of self-generated squeezing.

Both routes reach the same $\sqrt{\Gamma/N}$ floor — criticality buys gain and drift immunity, not a new scaling law. Adding Rydberg interactions to Route I bends the threshold back into optical bistability, which shows that the incumbent transition and the proposed one are two limits of a single model, with the superradiant solid as the coexistence region.

Contents

1	Same protocol, different phase transition	2
2	The model	3
2.1	Level scheme and the dressed sensor spin	3
2.2	Two ways to make a light–matter transition	4
3	Route I: the $U(1)$ (lasing) transition	4
3.1	From Zhang’s condensation to a lasing threshold	4
3.2	Threshold	4

3.3	The frequency read-out, and why it answers the drift objection	5
3.4	Sensitivity floor	6
4	Route II: the $\mathbb{Z}_2$ Dicke transition	6
4.1	Threshold line and order parameter	7
4.2	Three read-out protocols	7
4.3	Metrological analysis	8
4.3.1	Closed system: quantum Fisher information at criticality	8
4.3.2	Why the $N^{4/3}$ is not free	9
4.3.3	Open system: gain, bandwidth, squeezing and the input-referred floor	9
4.3.4	Sensitivity in physical units	11
5	Rydberg interactions: the link to the superradiant solid	11
5.1	Interactions bend Route I back into the incumbent transition	12
6	The three transitions compared	13
7	Prior art	14
7.1	Cavity-enhanced Rydberg electrometry — the direct lineage	14
7.2	Critical Rydberg metrology — the closest competitor	15
7.3	Critical quantum metrology — the theory that constrains the claims	15
7.4	Dicke-model engineering and Rydberg cavity QED — the feasibility base	15
7.5	What appears to be unoccupied	16
8	Assessment	16
8.1	Is it a good idea?	16
8.2	Benefits	17
8.3	Risks and objections	17
8.4	Recommended programme	18
A	Numerical methods	18

1 Same protocol, different phase transition

The proposal is best stated as a substitution rather than an invention.

Ding *et al.* [6] and Wang *et al.* [7] have established a sensing protocol: bias a Rydberg ensemble at a critical point, let the microwave field push the system across it, and read the steep optical response. It works — 49 and 2.6 nV cm⁻¹Hz^{-1/2} respectively. The critical point they use is that of a *non-equilibrium phase transition* (NEPT): the optical bistability of a laser-driven, interacting Rydberg gas, arising from the competition between long-range Rydberg interactions, drive and dissipation. Its order parameter is the Rydberg population, observed in transmission. The light is a probe; delete the cavity and the transition survives.

This note keeps the protocol and changes the transition. We use a **light–matter** transition, in which the cavity field is itself the order parameter — delete the cavity and there is no transition at all. The reason this is a sensor is one equation. In the Rydberg-EIT ladder the microwave field dresses the two Rydberg states and thereby fixes the splitting $\tilde{\omega}_0$ of the effective two-level system the cavity sees,

$$\tilde{\omega}_0 = \delta + \Omega_\mu/2, \quad \Omega_\mu = dE_\mu/\hbar, \quad (1)$$

and every light–matter phase boundary depends on $\tilde{\omega}_0$. *The boundary is therefore a function of the field to be measured.*

Two flavours of light–matter transition are available, and they differ in how much new physics they demand.

	NEPT (incumbent) Ding 2022, Wang 2025	Route I: $U(1)$ after Zhang 2013	Route II: $\mathbb{Z}_2$ after Dimer 2007 / Baden 2014
Driven by	Rydberg interactions	light–matter coupling (RWA)	light–matter coupling (+ counter-rotating)
Order parameter	Rydberg population	phase of emitted field	cavity field $\langle a \rangle$
Broken symmetry	none (Ising-like jump)	$U(1)$	$\mathbb{Z}_2$
Realisation	optical bistability	superradiant lasing threshold	ground-state QPT
Extra hardware	—	repump	second Raman channel

Table 1: The substitution. Sections 3 and 4 work out the two new columns; Sec. 6 compares all three.

Route I (Sec. 3) is the transition of Zhang *et al.* [2] — rotating-wave, $U(1)$ -breaking. Their calculation is at equilibrium, at fixed chemical potential; a driven optical cavity has no chemical potential, so the same broken symmetry is realised instead as a *superradiant lasing* threshold. This is the cheaper route: it adds only a repump to apparatus that already exists [31]. Its distinctive payoff is that in the bad-cavity limit the emitted frequency locks to the atoms, so the measurand is read as an optical frequency and cavity drift is rejected.

Route II (Sec. 4) is the $\mathbb{Z}_2$ Dicke transition, which needs engineered counter-rotating terms [11, 12, 29]. It is the more demanding experiment and the stronger physics claim: a genuine ground-state critical point, with divergent transduction gain and self-generated squeezing.

Section 2 sets up the shared front end, Secs. 3 and 4 work out the two routes, Sec. 5 puts the Rydberg interactions back in and shows that they bend Route I into the incumbent NEPT, Sec. 6 weighs the three against each other, Sec. 7 reviews what has been done, and Sec. 8 assesses.

A note on sourcing. The primary sources for every claim of the form “paper X does not do Y” in Sec. 7 have been read first-hand: Refs. [1, 3, 4] (abstracts and stated results), and Refs. [2, 6, 7] in full. One substantive correction came out of that reading and is recorded in Sec. 3: the model of Ref. [2] is written in the rotating-wave approximation, so its transition is $U(1)$ rather than the $\mathbb{Z}_2$ one, which is why Route I and Route II are separated here rather than conflated.

2 The model

2.1 Level scheme and the dressed sensor spin

Take the standard Rydberg-EIT electrometry ladder in ^{87}Rb : $|g\rangle = |5S_{1/2}\rangle$, $|e\rangle = |5P_{3/2}\rangle$, $|r\rangle = |53D_{5/2}\rangle$, $|r'\rangle = |54P_{3/2}\rangle$. A probe field addresses $|g\rangle \rightarrow |e\rangle$, a strong control laser Ω_c drives $|e\rangle \rightarrow |r\rangle$, and the microwave field to be measured drives $|r\rangle \rightarrow |r'\rangle$ with Rabi frequency $\Omega_\mu = dE_\mu/\hbar$. For this transition $d = 1774.82 ea_0$, giving the conversion

$$\Omega_\mu/2\pi = 2.271 \text{ MHz} \times (E_\mu/\text{mV cm}^{-1}). \quad (2)$$

On microwave resonance the Rydberg pair is dressed into $|\pm\rangle = (|r\rangle \pm |r'\rangle)/\sqrt{2}$ at energies $\pm\Omega_\mu/2$: this is the Autler–Townes doublet that conventional electrometry reads out as a line splitting. The key move here is to treat one dressed state, say $|+\rangle$, together with $|g\rangle$ as a single effective spin- $\frac{1}{2}$ whose splitting is

$$\tilde{\omega}_0 = \delta + \Omega_\mu/2, \quad (3)$$

δ being the two-photon detuning of the optical fields. *The measurand has become a Hamiltonian parameter of a two-level system*, not a spectroscopic feature to be resolved.

2.2 Two ways to make a light–matter transition

A collection of N atoms coupled to a cavity in the rotating-wave approximation,

$$H_{\text{TC}} = \omega a^\dagger a + \tilde{\omega}_0 b^\dagger b + G(a^\dagger b + a b^\dagger), \quad G = g_0 \sqrt{N}, \quad (4)$$

has *no* phase transition on its own: it is two coupled linear oscillators, and what one observes is the collective Rabi splitting of Refs. [3, 4]. Getting a transition out of light and matter requires adding one of two things.

Add gain (Route I). Incoherently repump the atoms. Equation (4) then acquires a threshold at which the collective gain balances the cavity loss, above which the field acquires a spontaneously chosen phase — a $U(1)$ -breaking superradiant lasing transition. This is the driven-dissipative counterpart of the polariton condensation of Ref. [2], whose equilibrium version is reached instead by tuning a chemical potential. Hardware cost: a repump beam.

Add counter-rotating terms (Route II). Restore $a^\dagger b^\dagger + ab$ to Eq. (4) and the model becomes the Dicke model, with a $\mathbb{Z}_2$ symmetry-breaking transition already in the *ground state*. A bare dipole coupling cannot do this — the A^2 term and the Thomas–Reiche–Kuhn sum rule forbid it [15, 16] — so the terms must be engineered, via the cavity-assisted Raman scheme of Dimer, Estienne, Parkins and Carmichael [11], realised by Baden *et al.* [12]. Reference [2] makes the complementary point that in a two-photon excitation scheme, where $|g\rangle$ is not directly dipole-coupled to the Rydberg state, the no-go theorem does not apply in the first place. Most directly, Ma *et al.* [29] give an explicit periodic-driving recipe for the *anisotropic* Dicke model in a cavity-coupled Rydberg array, with the ratio of counter-rotating to rotating terms tunable from zero to infinity. That is precisely the knob Route II needs, and it is already published. Hardware cost: a second Raman channel.

In both cases the measurand enters only through $\tilde{\omega}_0$, which is what makes Eq. (1) a sensing curve.

3 Route I: the $U(1)$ (lasing) transition

3.1 From Zhang’s condensation to a lasing threshold

Reading Ref. [2] in full turns up a point that shapes this whole section. Their Hamiltonian couples the cavity as $(g/\sqrt{N}) \sum_i (b_i^\dagger a_i \psi + \text{H.c.})$ — a rotating-wave coupling that conserves the total excitation number $N_{\text{ex}} = \psi^\dagger \psi + \sum_i n_i$, which is why their phase diagram is drawn against a chemical potential $\tilde{\mu}$ conjugate to N_{ex} . Their “generalised Dicke model” is therefore of Tavis–Cummings type, and their superradiance is a $U(1)$ -breaking polariton condensation, not the $\mathbb{Z}_2$ transition of Sec. 4.

An optical cavity has no chemical potential; it has pump and loss. The same broken symmetry is therefore realised in a driven system as a *superradiant lasing* threshold. The universality class differs from Zhang’s equilibrium condensation — this is a non-equilibrium, gain-driven transition — but the symmetry, the order parameter (the phase of the emitted field) and, crucially for us, the dependence of the boundary on $\tilde{\omega}_0$ all carry over.

3.2 Threshold

Semiclassical Maxwell–Bloch equations for a homogeneously broadened collective transition, in a frame rotating at the emission frequency ν :

$$\dot{a} = -(\kappa + iD_c) a + \frac{G^2}{g} p, \quad \dot{p} = -(\Gamma + iD_a) p + g a s, \quad \dot{s} = -\gamma_{\parallel}(s - s_0) - 2g(a^* p + \text{c.c.}), \quad (5)$$

with $D_c = \omega_{\text{cav}} - \nu$, $D_a = \tilde{\omega}_0 - \nu$, transverse decay Γ (including repump broadening), and collective gain $G^2 = g^2 N s_0 / 2$. Linearising about $a = p = 0$ gives $G^2 = (\kappa + iD_c)(\Gamma + iD_a)$. The imaginary

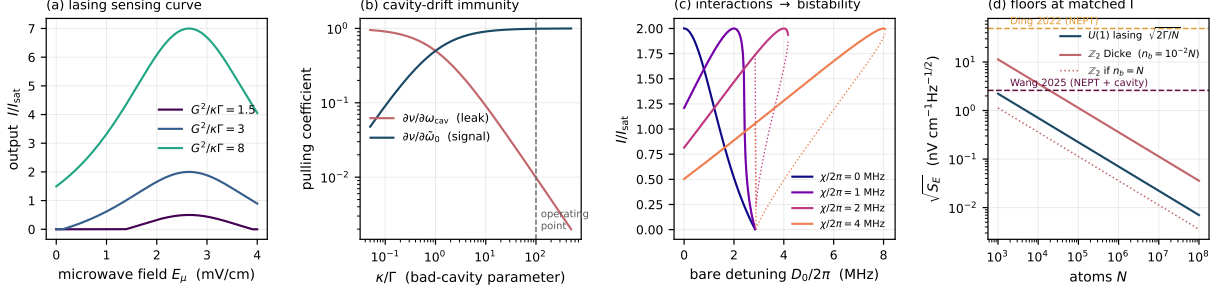


Figure 1: Route I. $\kappa/2\pi = 2$ MHz, $\Gamma/2\pi = 20$ kHz. (a) Output power versus microwave field: the field tunes the atom–cavity detuning through resonance, switching the laser on and off. (b) Cavity-pulling coefficients. In the bad-cavity limit the emission frequency follows the atoms ($\partial\nu/\partial\tilde{\omega}_0 \rightarrow 1$) and ignores the cavity ($\partial\nu/\partial\omega_{\text{cav}} \rightarrow \Gamma/\kappa$): at the operating point cavity drift is rejected $101\times$. (c) With a Rydberg-interaction shift χ pulling the transition towards resonance, the threshold folds over into bistability (dotted = unstable branch; $I = 0$ remains a solution throughout the folded region) — this is the incumbent NEPT, reached from the lasing side. (d) Quantum-noise floors at matched Γ , against the two published NEPT results.

part fixes the emission frequency and the real part the threshold. Writing $D \equiv \omega_{\text{cav}} - \tilde{\omega}_0$ for the atom–cavity detuning,

$$\nu = \frac{\kappa \tilde{\omega}_0 + \Gamma \omega_{\text{cav}}}{\kappa + \Gamma} \quad (\text{cavity pulling}), \quad (6)$$

$$G_{\text{th}}^2 = \kappa \Gamma \left[1 + \frac{D^2}{(\kappa + \Gamma)^2} \right] \quad (\text{threshold}). \quad (7)$$

We verified Eq. (7) against the largest real eigenvalue of the linearised (a, p) system: the two agree to all 8 digits printed, at $D/2\pi = 0, 1, 3$ and 8 MHz.

Since $D = \omega_{\text{cav}} - \delta - \Omega_\mu/2$, *the microwave field moves the lasing threshold*. At fixed collective gain the boundary is crossed at $|D_{\text{th}}| = (\kappa + \Gamma) \sqrt{G^2/\kappa\Gamma - 1}$; equivalently, servoing the two-photon detuning to hold the system at threshold gives

$$\delta_{\text{th}} = \omega_{\text{cav}} \mp (\kappa + \Gamma) \sqrt{\frac{G^2}{\kappa\Gamma} - 1} - \frac{\Omega_\mu}{2}, \quad \frac{\partial \delta_{\text{th}}}{\partial E_\mu} = -\frac{d}{2\hbar}, \quad (8)$$

the same SI-traceable scale factor as Autler–Townes electrometry. Figure 1(a) shows the resulting sensing curve.

3.3 The frequency read-out, and why it answers the drift objection

Equation (6) is the reason to prefer this route. The emission frequency is a weighted mean of the atomic and cavity frequencies, and in the bad-cavity limit $\kappa \gg \Gamma$ the atomic weight dominates:

$$\frac{\partial \nu}{\partial \tilde{\omega}_0} = \frac{\kappa}{\kappa + \Gamma} \rightarrow 1, \quad \frac{\partial \nu}{\partial \omega_{\text{cav}}} = \frac{\Gamma}{\kappa + \Gamma} \rightarrow \frac{\Gamma}{\kappa} \ll 1. \quad (9)$$

At $\kappa/2\pi = 2$ MHz and $\Gamma/2\pi = 20$ kHz we find 0.9901 and 9.90×10^{-3} : the signal is transferred essentially in full while cavity-length noise is suppressed $101\times$. The measurand appears as

$$\frac{\partial \nu}{\partial E_\mu} = \frac{\kappa}{\kappa + \Gamma} \cdot \frac{d}{2\hbar} = 1.124 \text{ MHz per (mV/cm)}, \quad (10)$$

i.e. 0.990 times the half-Autler–Townes slope — read out as an *optical frequency*, by beating against a reference.

This is the substantive advantage over both the incumbent NEPT and Route II. The standing objection to critical sensing is that a divergent susceptibility amplifies drift as readily as signal;

formally, the critical Fisher-information matrix is sloppy [32]. Here the dominant technical channel — cavity length — is not amplified but *actively rejected*, by the same mechanism that makes bad-cavity (“active optical clock”) lasers insensitive to their resonators. The rejection is a design parameter: it is κ/Γ , tunable by choosing the cavity finesse.

3.4 Sensitivity floor

Above threshold each repumped atom yields one cavity photon, $2\kappa n = Nw/2$, and the bad-cavity laser linewidth is the Schawlow–Townes value reduced by the squared pulling coefficient,

$$\Delta\omega = \frac{\kappa}{2n} \left(\frac{\Gamma}{\kappa + \Gamma} \right)^2 \xrightarrow{\kappa \gg \Gamma, w=\Gamma} \frac{2\Gamma}{N}. \quad (11)$$

Phase diffusion at rate $\Delta\omega$ gives $\sqrt{S_\nu} = \sqrt{\Delta\omega}$, so referring through the pulling coefficient and through $\tilde{\omega}_0 = \delta + \Omega_\mu/2$,

$$\boxed{\sqrt{S_{\tilde{\omega}_0}} = \frac{\kappa + \Gamma}{\kappa} \sqrt{\frac{2\Gamma}{N}}} \quad (12)$$

— the collective-atomic standard quantum limit, with all N atoms participating. Note this is a derivation of the floor, not an independent check of it: the two expressions are the same formula. In physical units:

N	intracavity photons	linewidth	$\sqrt{S_E}$ (nV cm ⁻¹ Hz ^{-1/2})	Route II at $n_b = 10^{-2}N$
10^4	25	3.9 Hz	0.703	3.58
10^5	250	392 mHz	0.222	1.13
10^6	2.5×10^3	39 mHz	0.070	0.358
10^7	2.5×10^4	3.9 mHz	0.022	0.113
10^8	2.5×10^5	0.39 mHz	0.0070	0.036

Table 2: Route I frequency-pulling read-out, $\kappa/2\pi = 2$ MHz, $\Gamma/2\pi = 20$ kHz, repump $w = \Gamma$. The last column is Route II’s floor at the same N and Γ (Sec. 4), which is capped by the Holstein–Primakoff requirement $n_b \ll N$.

Route I is a factor 5.1 better at equal atom number, for a structural reason worth stating plainly: the lasing route inverts and uses *all* N atoms, whereas Route II linearises about the unexcited state and is limited to $n_b \ll N$ collective excitations. The mHz linewidths are the standard superradiant-laser prediction and are the reason the frequency read-out is competitive at all.

4 Route II: the $\mathbb{Z}_2$ Dicke transition

Restoring the counter-rotating terms to Eq. (4) turns the Tavis–Cummings model into the Dicke model,

$$H = \omega a^\dagger a + \tilde{\omega}_0 J_z + \frac{2\lambda}{\sqrt{N}} (a + a^\dagger) J_x, \quad (13)$$

with J_α the collective spin operators in the $j = N/2$ manifold and λ the engineered coupling of Sec. 2.2. Two features matter for sensing. First, ω and $\tilde{\omega}_0$ are *effective* frequencies — detunings in a rotating frame — so they can be dialled from kHz to MHz and reaching $\lambda \sim \sqrt{\omega\tilde{\omega}_0}/2$ does not require ultrastrong coupling. Second, $\lambda \propto \Omega_{\text{Raman}} g_0 \sqrt{N}/\Delta_e$ is set by laser power and is tunable *in situ*, so the operating point can be servoed onto threshold.

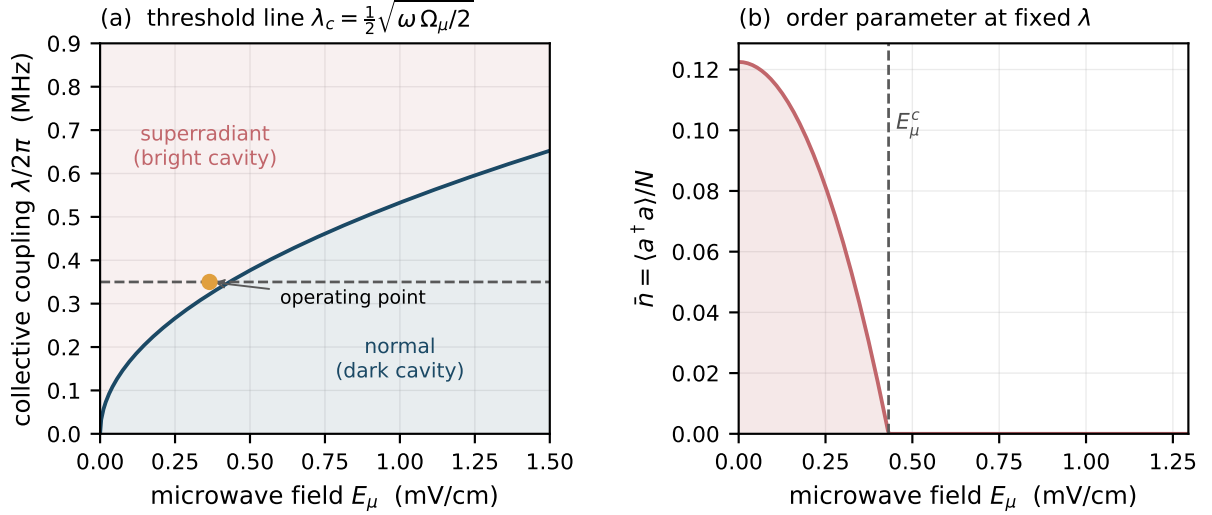


Figure 2: The microwave field moves the phase boundary. (a) Threshold coupling $\lambda_c = \frac{1}{2}\sqrt{\omega \Omega_\mu/2}$ versus microwave field for $\omega/2\pi = 1$ MHz, $\delta = 0$ and the $53D_{5/2} \rightarrow 54P_{3/2}$ dipole. At fixed λ (dashed) the system crosses from superradiant to normal at a well-defined field. (b) Order parameter $\bar{n}$ at $\lambda/2\pi = 0.35$ MHz: the cavity goes dark above a critical field E_μ^c . The steep square-root onset near E_μ^c is the transduction gain the scheme trades on.

4.1 Threshold line and order parameter

Mean-field minimisation of Eq. (13) reproduces the textbook result [14]. Writing $\langle J_z \rangle = -\frac{N}{2} \cos \theta$, $\langle J_x \rangle = \frac{N}{2} \sin \theta$ and $\alpha = \langle a \rangle / \sqrt{N}$,

$$\frac{E}{N} = \omega \alpha^2 - \frac{\tilde{\omega}_0}{2} \cos \theta + 2\lambda \alpha \sin \theta, \quad \alpha = -\frac{\lambda}{\omega} \sin \theta, \quad (14)$$

which has the normal solution $\theta = 0$ for $\lambda < \lambda_c$ and, for $\lambda > \lambda_c$,

$$\cos \theta = \mu \equiv \frac{\lambda_c^2}{\lambda^2} = \frac{\omega \tilde{\omega}_0}{4\lambda^2}, \quad \bar{n} \equiv \frac{\langle a^\dagger a \rangle}{N} = \frac{\lambda^2}{\omega^2} (1 - \mu^2), \quad \lambda_c = \frac{1}{2} \sqrt{\omega \tilde{\omega}_0}. \quad (15)$$

Figure 2 shows the resulting sensing curve and the order parameter as the microwave field is swept at fixed λ .

The immediate observation is that the *sign* of the response is the useful one: for $\delta = 0$ a *larger* microwave field raises $\tilde{\omega}_0$ and hence λ_c , extinguishing superradiance. The sensor is therefore naturally operated as a bright-to-dark discriminator, which is favourable for detection (one measures a decrease from a large signal, not an increase from zero).

4.2 Three read-out protocols

The construction supports three distinct protocols, with quite different characters. It is worth separating them because most of the confusion in the critical-metrology literature comes from conflating them.

Protocol A — order-parameter read-out (analogue). Bias at fixed λ slightly above threshold and monitor the emitted light. The signal is $\partial \alpha / \partial \tilde{\omega}_0$, which diverges as $\mu \rightarrow 1$:

$$\frac{\partial \alpha}{\partial \tilde{\omega}_0} = \frac{\mu}{4\lambda \sqrt{1 - \mu^2}} \xrightarrow{\mu \rightarrow 1} \infty. \quad (16)$$

This is the direct analogue of what Ding *et al.* [6] and Wang *et al.* [7] do with the Rydberg-interaction-driven critical point, and it is the protocol most likely to be implemented first. Its

weakness is that the divergent susceptibility is not selective: laser intensity noise (which moves λ), cavity length drift (which moves ω) and stray dc fields (which move δ) are amplified by exactly the same factor as the signal.

Protocol B — threshold (null) read-out. Hold λ and ω fixed and servo the two-photon detuning δ so that the system sits exactly at threshold. The lock point satisfies

$$\delta_{\text{th}} = \frac{4\lambda^2}{\omega} - \frac{\Omega_\mu}{2} \implies \frac{\partial \delta_{\text{th}}}{\partial E_\mu} = -\frac{d}{2\hbar}, \quad (17)$$

so a change in the microwave field is read out directly as a change in an *optical laser detuning*, which is a frequency and is therefore SI-traceable in exactly the sense that makes Autler–Townes electrometry attractive [9, 10]. The scale factor $d/2\hbar$ is the same as for Autler–Townes; what changes is the discriminant. In AT electrometry the resolution is set by how well a splitting can be resolved against the EIT linewidth. Here it is set by how sharply the threshold can be located, which is a different — and, near a continuous transition, potentially much steeper — function.

To our reading of the literature this null variant has not been proposed, and it is the most defensible original element of the idea. It also has a practical virtue: null methods are first-order insensitive to gain calibration, which is precisely the weakness of Protocol A.

Protocol C — critical superheterodyne (recommended). Apply a microwave local oscillator with Rabi frequency Ω_{LO} to set the bias point $\tilde{\omega}_0 = \delta + \Omega_{\text{LO}}/2$ just below threshold, and let the weak signal field at $\omega_{\text{LO}} + \omega_{\text{IF}}$ beat against it:

$$\tilde{\omega}_0(t) = \delta + \frac{\Omega_{\text{LO}}}{2} + \frac{\Omega_{\text{s}}}{2} \cos(\omega_{\text{IF}}t). \quad (18)$$

The measurand is now a small *modulation* of a Hamiltonian parameter, amplified by the near-threshold susceptibility and detected at the intermediate frequency. This inherits the linear-in- E_{s} scaling and technical-noise rejection that make the Rydberg superheterodyne receiver [8] the most sensitive Rydberg electrometer built, and adds critical gain on top. It is the configuration used for all the open-system numbers in Sec. 4.3.3, and it is the one we would actually build.

4.3 Metrological analysis

4.3.1 Closed system: quantum Fisher information at criticality

We first ask what the *ground state* of Eq. (13) is worth as a probe of $\tilde{\omega}_0$. Exact diagonalisation in the $j = N/2$ manifold (even-parity sector; photon cutoff converged to 10^{-12} , see Appendix A) gives the quantum Fisher information from the fidelity susceptibility, $F_Q = 8(1 - \langle \psi(\tilde{\omega}_0) | \psi(\tilde{\omega}_0 + h) \rangle) / h^2$.

Figure 3 confirms the expected fully-connected-model exponents [18]: the gap closes as $N^{-1/3}$ and

$$F_Q[\tilde{\omega}_0] \big|_{\lambda=\lambda_c} \propto N^{4/3}, \quad (19)$$

with a fitted exponent of 1.307 (asymptotic value $4/3 = 1.333$; the residual is the expected slow finite-size correction, the fit drifting monotonically upward as the window moves to larger N). In the normal phase we find, at $N = 512$ and in the window $N^{-2/3} \ll \varepsilon \ll 1$,

$$F_Q[\tilde{\omega}_0] \propto \varepsilon^{-2}, \quad \varepsilon \equiv 1 - \lambda^2/\lambda_c^2, \quad (20)$$

with a fitted exponent of -2.0008 — the Gaussian critical scaling of a squeezed soft mode, cut off at $\varepsilon_{\text{min}} \sim N^{-2/3}$, which is exactly what produces the $N^{4/3}$ peak.

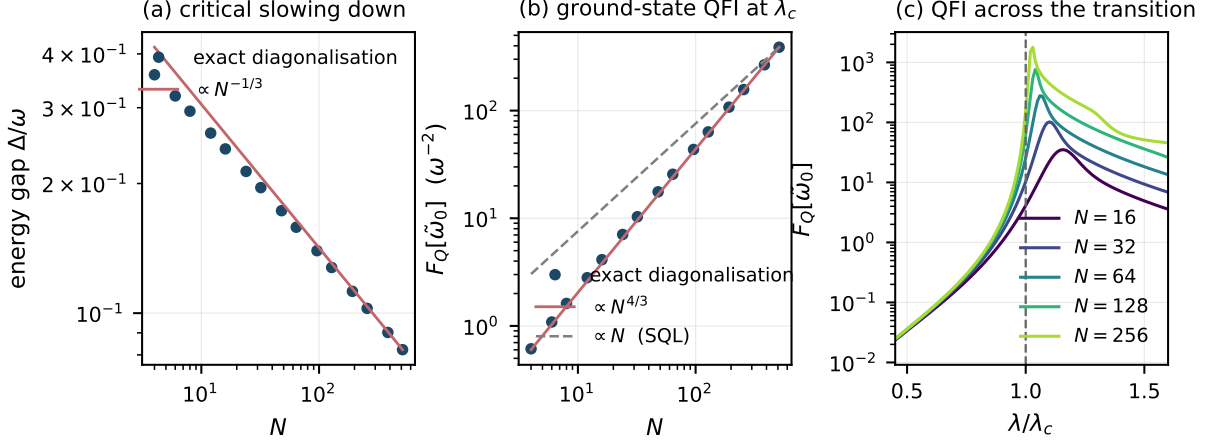


Figure 3: Finite-size scaling at the critical point ($\omega = \tilde{\omega}_0 = 1$, $\lambda = \lambda_c$, $N \leq 512$). (a) The energy gap closes as $N^{-1/3}$ (fitted -0.317 over $N \geq 128$). (b) The quantum Fisher information for $\tilde{\omega}_0$ grows as $N^{4/3}$ (fitted 1.307 over $N \geq 128$), super-extensive and thus formally beyond the standard quantum limit $F_Q \propto N$ (dashed). (c) F_Q across the transition; the peak sharpens and moves towards λ_c as N grows.

4.3.2 Why the $N^{4/3}$ is not free

Taken at face value, $F_Q \propto N^{4/3}$ says $\delta\Omega_\mu \propto N^{-2/3}$, beating the $N^{-1/2}$ standard quantum limit. This is the claim that critical-metrology proposals are often built on, and it does not survive proper resource accounting. The gap closes as $N^{-1/3}$, so adiabatic preparation of the critical ground state takes a time $T_{\text{prep}} \propto N^{1/3}$. Converting to a per-root-Hertz sensitivity,

$$\delta\Omega_\mu \sqrt{T_{\text{prep}}} \propto N^{-2/3} \cdot N^{1/6} = N^{-1/2}, \quad (21)$$

and the advantage cancels exactly. This is the content of Rams *et al.* [19]; Gietka *et al.* [20] show the cancellation survives shortcuts to adiabaticity, and Gyhm *et al.* [22] have recently established a general T^6 ceiling on the quantum Fisher information of monotonic critical protocols. A proposal built on the $N^{4/3}$ scaling alone would be building on sand, and this document should not be used to write one.

The interesting question is therefore not the scaling but the *prefactor*, which is what the driven open system delivers.

4.3.3 Open system: gain, bandwidth, squeezing and the input-referred floor

We linearise Eq. (13) in the normal phase (Holstein–Primakoff bosons a for the cavity, b for the collective atomic mode), add cavity decay κ and collective decoherence γ , and drive the cavity coherently. In quadratures $v = (x_a, p_a, x_b, p_b)$,

$$\dot{v} = Av + f + Bu, \quad A = \begin{pmatrix} -\kappa & \omega & 0 & 0 \\ -\omega & -\kappa & -2\lambda & 0 \\ 0 & 0 & -\gamma & \tilde{\omega}_0 \\ -2\lambda & 0 & -\tilde{\omega}_0 & -\gamma \end{pmatrix}, \quad (22)$$

with vacuum inputs u . The measurand enters as $\delta H = \delta\tilde{\omega}_0 b^\dagger b$, i.e. as a coherent force

$$s = \delta\tilde{\omega}_0 (0, 0, \bar{p}_b, -\bar{x}_b) \quad (23)$$

proportional to the steady-state collective amplitude — which is why Protocol C needs the probe drive. Output homodyne spectra follow from $x_{\text{out}} = \sqrt{2\kappa} x_a - x_{\text{in}}$. The implementation is validated by reproducing the exact vacuum level $S_{\text{out}} = 1/2$ for an empty cavity.

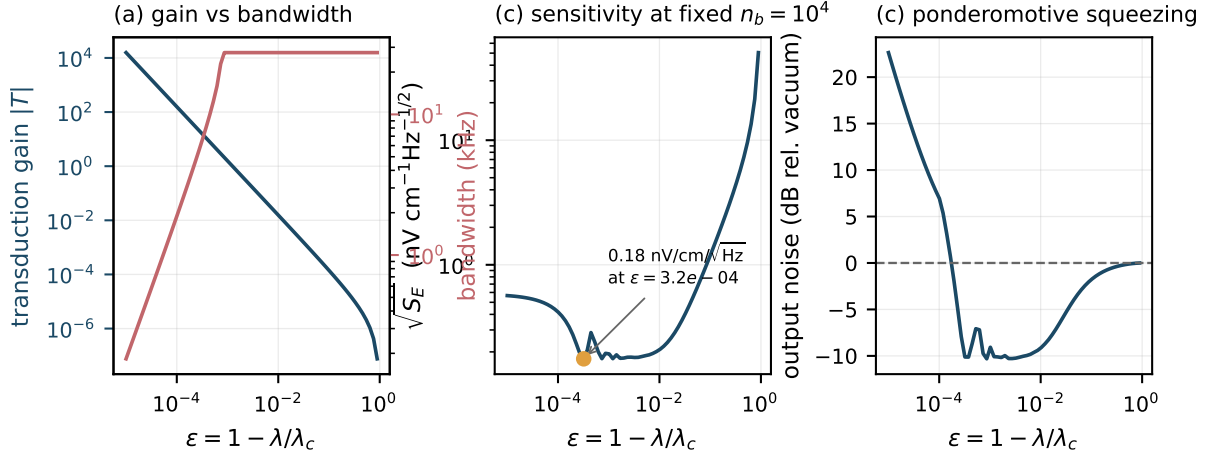


Figure 4: Driven, damped sensor near threshold. (a) Transduction gain (left axis) and response bandwidth (right axis) versus $\varepsilon = 1 - \lambda/\lambda_c$. At fixed collective occupation the gain grows as ε^{-1} throughout, while the bandwidth stays pinned at $(\kappa + \gamma)/2 = 27.5$ kHz until $\varepsilon \lesssim 3 \times 10^{-4}$ and only then collapses as ε^{+1} . There is thus a window of *free* gain before critical slowing down begins. (b) Quantum-noise-limited field sensitivity at fixed $n_b = 10^4$, with a broad optimum of $0.18 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$. (c) The near-threshold system squeezes its own output by up to 10.3 dB before atomic noise takes over.

Throughout we use $\omega/2\pi = \tilde{\omega}_0/2\pi = 1$ MHz, $\kappa/2\pi = 50$ kHz and $\gamma/2\pi = 5$ kHz, giving a damped threshold $\lambda_c/2\pi = 0.5006$ MHz.

Figure 4 summarises the behaviour. Three points are worth stating precisely.

Gain and bandwidth. At fixed collective occupation $n_b = \langle b^\dagger b \rangle$, the transduction gain scales as $\varepsilon^{-1.000}$ over six decades. The bandwidth, however, is *not* immediately paid for: the slowest relaxation rate stays pinned at $(\kappa + \gamma)/2 = 27.5$ kHz until $\varepsilon \lesssim 3 \times 10^{-4}$, below which the soft mode becomes overdamped and the rate falls as $\varepsilon^{+1.002}$, giving a strict gain–bandwidth product (constant to 0.78%). Practically: there is roughly three decades of ε over which criticality buys 10^3 in gain at no bandwidth cost, and beyond that it is a straight trade. This is a genuinely useful design statement and it does not appear to have been made for this system.

Intrinsic squeezing. Near threshold the soft mode is a degenerate parametric amplifier and the output is squeezed — we find up to -10.3 dB at $\varepsilon \approx 9 \times 10^{-4}$, degrading to excess noise once $\varepsilon \lesssim 10^{-4}$ where atomic decoherence dominates. The sensor generates its own squeezed light; no external squeezed source is needed. This is a real, if modest, quantum-enhancement channel and it is compatible with Protocol C.

The control experiment that settles the matter. Sensitivity figures near a critical point are easy to inflate, because approaching threshold increases the steady-state excitation as well as the gain. We therefore repeated the calculation holding n_b *fixed* by re-solving for the drive at each ε . The result is unambiguous:

$$\boxed{\sqrt{S_{\tilde{\omega}_0}} = 0.72 \sqrt{\gamma/n_b}} \quad (24)$$

with the numerical prefactor constant to one part in 10^4 across five decades of n_b and, away from the extreme critical region, across three decades of ε . Equation (24) is the standard quantum limit for n_b atoms with coherence time $1/\gamma$, up to a factor of order unity that we attribute to the intrinsic squeezing.

The physical reading: **criticality is a transducer, not a source of metrological advantage**. It converts the atomic signal into a large optical signal — which is exactly what one needs to beat photodetector and technical noise, and is why Refs. [6, 7] report large Fisher-information

gains — but it does not lower the floor set by how many atoms are coherently participating and for how long.

4.3.4 Sensitivity in physical units

Converting Eq. (24) through $\delta\tilde{\omega}_0 = \delta\Omega_\mu/2$ and $E_\mu = \hbar\Omega_\mu/d$:

Collective occupation n_b	$\sqrt{S_E}$ (nV cm ⁻¹ Hz ^{-1/2})	SQL for $N = n_b$	ratio
10 ³	0.567	0.786	0.72
10 ⁴	0.179	0.248	0.72
10 ⁵	0.057	0.079	0.72
10 ⁶	0.018	0.025	0.72
10 ⁷	0.0057	0.0079	0.72

Table 3: Quantum-noise-limited field sensitivity at $\varepsilon = 3 \times 10^{-3}$, $\gamma/2\pi = 5$ kHz. For reference: the Rydberg superheterodyne receiver of Ref. [8] achieves 55 nV cm⁻¹Hz^{-1/2}; Ref. [7] quotes 5.1 nV cm⁻¹Hz^{-1/2} as the best sensitivity from conventional SNR-enhancement methods, 49–50.3 nV cm⁻¹Hz^{-1/2} for free-space critical (non-equilibrium) electrometry, and 2.6 nV cm⁻¹Hz^{-1/2} for its own cavity-enhanced critical measurement.

Two caveats attach to every number in that table. First, Holstein–Primakoff validity requires $n_b \ll N$; taking $n_b/N \lesssim 10^{-2}$, the $n_b = 10^6$ row needs $N \gtrsim 10^8$ atoms coherently coupled to one cavity mode, which is demanding but not absurd for a vapour cell and out of reach for a small cold cloud. Second, these are *quantum-noise-limited* figures with no allowance for Doppler broadening, transit-time effects, inhomogeneous coupling across the cavity mode, laser frequency noise or dc-field drift, all of which are what actually limits real Rydberg electrometers. Treat them as an upper bound on what the mechanism can offer, not a prediction.

5 Rydberg interactions: the link to the superradiant solid

Reference [2] is not the plain Dicke model: its content is what happens when Rydberg–Rydberg interactions compete with the cavity coupling, producing a superradiant *solid* in which superradiance and crystalline order coexist, with first-order boundaries. It is worth asking whether those interactions help or hurt the sensor.

Adding a nearest-neighbour Rydberg repulsion to Eq. (13),

$$H_{\text{DI}} = H + V \sum_{\langle ij \rangle} n_i n_j, \quad n_i = S_i^z + \frac{1}{2}, \quad (25)$$

and taking a two-sublattice mean field on a bipartite lattice of coordination z (with α eliminated by its own stationarity condition),

$$\frac{E}{N} = \omega\alpha^2 - \frac{\tilde{\omega}_0}{4}(\cos\theta_A + \cos\theta_B) + \lambda\alpha(\sin\theta_A + \sin\theta_B) + \frac{Vz}{2}n_A n_B, \quad \alpha = -\frac{\lambda}{2\omega}(\sin\theta_A + \sin\theta_B), \quad (26)$$

reproduces the phase structure of Ref. [2]. We minimise this from 13×13 seeds per point with L-BFGS-B, and use adiabatic up/down sweeps in λ as the first-order diagnostic. Figure 5 shows the result, and it splits cleanly into two regimes.

At the sensor’s operating point, interactions are inert. For $\tilde{\omega}_0 > 0$ the normal-phase Rydberg population vanishes, so $n_A n_B \rightarrow 0$ and the interaction term does nothing *at* the transition. On a fine scan we find the onset at $\lambda/\omega = 0.5005$ for every $V \in \{0, 0.5, 2, 5\}$, i.e. no shift resolvable at the 5×10^{-4} level. Only the amplitude above threshold is suppressed (Fig. 5b). This is a genuinely useful robustness result: *the calibration curve Eq. (1) is immune to Rydberg*

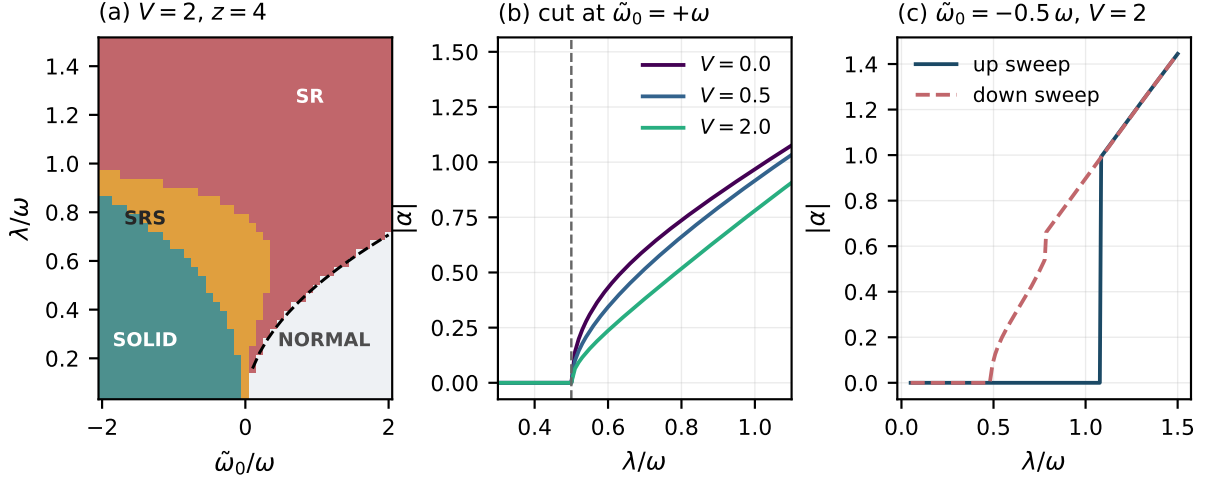


Figure 5: Rydberg interactions and the sensing threshold ($V = 2\omega$, $z = 4$). (a) Mean-field phase diagram. The superradiant solid (SRS) of Ref. [2] occupies only $\tilde{\omega}_0 < 0$; for $\tilde{\omega}_0 > 0$ — the sensor’s operating regime — the boundary is the plain Dicke line $\lambda_c = \frac{1}{2}\sqrt{\omega\tilde{\omega}_0}$ (dashed), unmodified. (b) Cuts at $\tilde{\omega}_0 = +\omega$ for $V = 0, 0.5, 2$: the onset does not move, only the post-threshold amplitude is suppressed. (c) At $\tilde{\omega}_0 = -0.5\omega$ the transition is strongly first order, with a hysteresis loop of height 0.985 in $|\alpha|$.

interactions, so atomic density fluctuations — which are exactly what moves the critical point in Refs. [6, 7] — do not move it here. It is the sharpest technical argument for preferring the Dicke critical point over the interaction-driven one.

The superradiant solid lives elsewhere, and it is first order. The SOLID and SRS phases appear only for $\tilde{\omega}_0 < 0$, i.e. when the dressed Rydberg level sits *below* $|g\rangle$ in the rotating frame (readily arranged by choosing $\delta < -\Omega_\mu/2$). There the sequence on increasing λ is SOLID $\rightarrow$ SRS $\rightarrow$ SR, and the transition is strongly first order: up- and down-sweeps differ by 0.985 in $|\alpha|$ at $\tilde{\omega}_0 = -0.5\omega$, $V = 2\omega$.

The design conclusion, however, is largely independent of the details and is worth stating now, because it runs against the intuition that "more Rydberg physics is better":

- **For Protocols A and C, Rydberg interactions are a nuisance.** The clean collective enhancement of the Dicke model assumes N independent emitters sharing one bright mode. Rydberg blockade caps the number of independently excitable atoms per blockade volume at one, so the effective N entering $\lambda = g_0\sqrt{N}$ and the floor (24) is $N_{\text{eff}} = V_{\text{mode}}/V_{\text{blockade}}$, not the atom number. Interactions also dephase the collective mode, raising γ . This is not a speculative worry: the quantum Monte Carlo study of Ref. [28] reports exactly that, finding that Rydberg blockade significantly suppresses the cavity occupation. The sensor wants *weak* interactions — moderate n , or a Rydberg pair chosen for small van der Waals coefficient.
- **For a threshold detector, they are a feature.** If the transition is driven first order, the order parameter jumps discontinuously and the system latches, with hysteresis. That is useless for analogue metrology but excellent for a *trigger*: a device that fires when the field crosses a settable level, with essentially unbounded gain at the switching point and built-in hysteresis to suppress chatter. This is a legitimate secondary application and it is the one place where Ref. [2]’s first-order physics is directly useful.

5.1 Interactions bend Route I back into the incumbent transition

The same question can be asked of Route I, and the answer there is more interesting. Rydberg interactions shift the dressed transition in proportion to the Rydberg population, i.e. to the

intracavity intensity, $D(I) = D_0 - \chi I$. Above threshold the gain clamps, so the steady state satisfies

$$I = \frac{G^2}{G_{\text{th}}^2(D_0 - \chi I)} - 1. \quad (27)$$

When the shift pulls the transition *towards* resonance the feedback is positive, and Eq. (27) acquires multiple roots. Solving it on a grid at $G^2 = 3\kappa\Gamma$ (so that the bare system lases only for $|D|/2\pi < 2.86$ MHz), we find a single-valued response for $\chi/2\pi \leq 1$ MHz and bistability — two lasing roots coexisting with the dark state $I = 0$ — for $\chi/2\pi \gtrsim 2$ MHz; see Fig. 1(c). At $\chi/2\pi = 2$ MHz and $D_0/2\pi = 4$ MHz the roots sit at $I = 1.61$ and 2.00 .

That bistability, with its dark and bright branches and its hysteresis, *is* the non-equilibrium transition Ding *et al.* and Wang *et al.* use. So the incumbent transition and Route I are not rivals from different worlds: they are two limits of one model. Interactions dominant $\Rightarrow$ optical bistability (the NEPT). Light–matter coupling dominant $\Rightarrow$ the lasing threshold (Route I). Both comparable $\Rightarrow$ the coexistence region, the driven-dissipative counterpart of Zhang’s superradiant solid.

This is a useful thing to be able to say in a paper: the proposal is not a competing claim about which transition is “the” right one, but a statement that the sensor should be operated in the light–matter-dominated corner of a phase diagram whose other corner is already known to work. It also sets the design rule — keep χ small (moderate n , or a Rydberg pair with a small van der Waals coefficient) unless the folded, latching response is wanted deliberately.

Neither Ref. [2] nor Ref. [28] addresses sensing; both are equilibrium phase-structure studies. Reference [30] is the closest existing work to Route I — a Rydberg medium in a cavity with a narrow transition, where the Rydberg nonlinearity “strongly modifies the superradiance threshold” — but its aim is generating ultranarrow nonclassical light, not measuring a field.

6 The three transitions compared

	NEPT (incumbent)	Route I ($U(1)$ lasing)	Route II ($\mathbb{Z}_2$ Dicke)
Scale factor	empirical, calibrated	$-d/2\hbar$ (exact)	$-d/2\hbar$ (exact)
Read-out	optical amplitude	optical <i>frequency</i>	amplitude / homodyne
Cavity-drift immu- nity	none	rejected $\kappa/\Gamma \approx 10^2$	none; drift amplified with signal
Critical point set by	density, interactions	repump, N , finesse	laser power (in situ, μs)
Reproducibility	drifts with T , density, stray fields	instrument setting	instrument setting
Hysteresis	yes (scan-direction dependent)	no (second order)	no (second order)
Noise floor	empirical: 49, 2.6	$\sqrt{2\Gamma/N}$, all N atoms	$0.72\sqrt{\Gamma/n_b}$, $n_b \ll N$
At $N = 10^6$, matched Γ	—	0.070	0.358
Transduction gain	steep edge	threshold nonlinearity	ε^{-1} , 3 decades free
Squeezing	no	no (classical light)	−10.3 dB, self-generated
Bandwidth	fast (vapour)	$\sim \kappa$	$(\kappa + \gamma)/2 \approx 27.5$ kHz
Extra hardware	none	repump	second Raman channel
Maturity	demonstrated twice	apparatus exists [31]	engineering published [29]

Table 4: The three critical mechanisms at matched atomic decoherence Γ . Sensitivities in $\text{nV cm}^{-1}\text{Hz}^{-1/2}$.

Where Route I wins. Cavity-drift immunity, and it is not a small effect: the dominant technical noise channel is suppressed by κ/Γ rather than amplified. It also uses every atom, giving a floor a factor 5.1 below Route II at equal N , and it reads the measurand out as an optical frequency — the most precisely measurable quantity there is. It is also the cheaper experiment.

Where Route II wins. It is a genuine ground-state critical point, so the apparatus of critical quantum metrology applies to it; it delivers ε^{-1} transduction gain with a wide window free of bandwidth cost; and it generates its own squeezed light (-10.3 dB), which Route I — producing essentially classical laser light — does not. If the goal is a physics result rather than an instrument, Route II is the stronger claim.

Where both lose to the incumbent. The NEPT has been *built*, twice, in a hot vapour cell. Both routes here need a good optical cavity and, for Route I, a narrow transition plus repumping; Route II most likely needs cold atoms. Neither offers a better scaling law: all three go as $1/\sqrt{N}$, and Sec. 4 shows explicitly that criticality does not move that exponent. What the light-matter routes offer is a better *prefactor* and, more importantly, a critical point that is an instrument setting rather than a property of the sample.

Recommendation. Route I first: cheaper, lower floor, and its drift immunity is the single most defensible advantage over the incumbent. Route II is the follow-up, and where the interesting quantum-metrology physics lives.

7 Prior art

The relevant literature falls into four strands. The purpose of this section is to be honest about how much of the idea is already occupied.

7.1 Cavity-enhanced Rydberg electrometry — the direct lineage

Peng and co-workers have published a coherent programme along exactly this line. Reference [1], the seed paper, places a four-level Rydberg-EIT medium in a cavity and reports a minimum detectable field roughly $8\times$ smaller than the cavity-free case and a spectral resolution improved by more than $20\times$, attributing both to cavity narrowing of the transmission features; it also notes robustness against control-field amplitude and broadband tunability. The same group then moved to the collective regime: Ref. [3] uses *collective Rabi splitting*, obtaining a four-peak cavity transmission whose outer splitting varies linearly with the microwave field, and reports about $7\times$ improvement over plain EIT; Ref. [4] reports $196.7\times$ over a bare atomic medium and $26.2\times$ over weak coupling, a minimum detectable strength of 396.5 nV cm^{-1} (a detection floor, not a per-root-hertz sensitivity), and resolution gains of $216\times$ and $10\times$. Experimentally, Ref. [5] demonstrates 18 dB of power-sensitivity enhancement using a microwave (not optical) resonator, and Ref. [31] demonstrates a cavity-enhanced Rydberg *superheterodyne* receiver with a 19 dB sensitivity gain.

Implication for the idea. The collective-coupling step is already taken. Anything proposed here must be distinguished from “ $g_0\sqrt{N}$ in a cavity”, because Ref. [3] is precisely that. The distinguishing feature has to be the *counter-rotating* terms and the phase transition they enable: on their published abstracts, Refs. [3, 4] are normal-phase Tavis–Cummings physics — a vacuum-Rabi/collective-Rabi splitting read out spectroscopically — with no transition and no critical point. Note this is exactly the same rotating-wave restriction found in Ref. [2] (Sec. 5), so the counter-rotating terms are the single ingredient that separates this proposal from the whole of that lineage.

7.2 Critical Rydberg metrology — the closest competitor

Ding, Liu, Shi, Guo, Mølmer and Adams [6] demonstrated many-body enhanced microwave metrology at the critical point of a non-equilibrium Rydberg gas, reporting a Fisher information three orders of magnitude above the independent-particle value and an equivalent sensitivity of $49 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$. Wang, Liang, Wang and co-workers [7] then put that critical system inside an optical cavity, reporting more than two orders of magnitude of additional Fisher information and $2.6 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$.

Both papers were read in full, and the mechanism is unambiguous. The critical point is that of a *non-equilibrium phase transition* (NEPT) — the optical bistability of a laser-driven Rydberg ensemble, arising from the competition between long-range Rydberg interactions, drive strength and dissipation. The “critical point” is operationally the coupling-laser frequency at which the transmission jumps, the jump direction depends on the scan direction, and a hysteresis loop is observed. Reference [7] works in a hot caesium vapour cell ($6S_{1/2} \rightarrow 6P_{3/2} \rightarrow 47D_{5/2}$, microwave to $48P_{3/2}$), uses the sharp bistability edge as a frequency discriminator, and quantifies it with the *classical* Fisher information; the cavity’s role is to steepen that edge (by $> 10\times$ in slope, $> 100\times$ in Fisher information). No light–matter phase transition is involved anywhere in either work.

This is the strongest prior art and it must be addressed head-on. Reference [7] is “critical Rydberg metrology in a cavity”, which is a fair one-line summary of the present idea too. The substantive difference is *which* criticality:

- Refs. [6, 7] use the *dissipative, Rydberg-interaction-driven* phase transition (optical bistability of the driven Rydberg ensemble). Its critical point is set by atomic density and interaction strength, and therefore drifts with temperature, density and stray fields, and cannot be tuned quickly. It is also hysteretic, so the discriminator is scan-direction dependent.
- The present proposal uses the *light-matter* (Dicke) transition. Its critical point is set by $\lambda \propto$ laser amplitude and by ω , both of which are tunable in situ on microsecond timescales and can be servoed. The critical point is a controlled instrument setting rather than a property of the sample.

That difference is real and is worth a paper, but it is a difference of *controllability and traceability*, not of sensitivity scaling. Claiming the latter would be wrong (see Sec. 4.3.3).

7.3 Critical quantum metrology — the theory that constrains the claims

Garbe *et al.* [17] established the finite-component critical metrology paradigm with the quantum Rabi model, and Ref. [18] worked out the fully-connected-model scaling (static, quench and finite-time-ramp protocols, including Kibble–Zurek scaling) whose exponents we reproduce in Fig. 3. Against these, Rams *et al.* [19], Gietka *et al.* [20, 21] and Gyhm *et al.* [22] constrain what can be claimed once time is counted as a resource. Any write-up of this idea must engage with that second group of references explicitly; a referee will otherwise do it for you.

7.4 Dicke-model engineering and Rydberg cavity QED — the feasibility base

The engineered Dicke model is experimentally established: proposed by Dimer *et al.* [11], realised by Baden *et al.* [12] with cavity-assisted Raman transitions, and separately by Baumann *et al.* [13] via self-organisation of a BEC. For the Rydberg platform specifically, Ref. [29] gives a periodic-driving recipe for the anisotropic Dicke model in a cavity-coupled Rydberg array with a counter-rotating/rotating ratio tunable from zero to infinity, and Ref. [28] maps the resulting phase diagram by quantum Monte Carlo. The enabling theory is therefore in place; what is missing is the sensing application. Cavity Rydberg polaritons have been observed [23], including the suppression of inhomogeneous broadening by collective strong coupling that this proposal needs. On the sensing side, Rydberg electrometers now approach the standard quantum limit in cold atoms [24] and reach within 1.0 dB of it in tweezer arrays [25], and collective-radiative linewidth

engineering for Rydberg microwave sensing has just been proposed [26]. Rydberg systems can also be spin-squeezed directly [27], which is the main competing route to sub-SQL electrometry.

7.5 What appears to be unoccupied

An arXiv full-text search combining *Dicke/superradiant/lasing threshold/cavity pulling/bad cavity* with *Rydberg, electrometry, microwave electric field, metrology* and *sensing* returns no hit that uses a light-matter transition as the sensing element for microwave electrometry. The Dicke–Rydberg literature (Refs. [2, 29, 28, 35]) is uniformly concerned with phase structure and dynamics; the critical-electrometry literature (Refs. [6, 7]) is uniformly concerned with the interaction-driven NEPT; and Ref. [30], which does combine a Rydberg medium with a superradiance threshold in a cavity, aims at generating nonclassical light. On that basis:

1. **Unoccupied.** Using a *light-matter* transition — either flavour — as the critical resource for microwave electrometry, in place of the Rydberg-interaction transition of Refs. [6, 7].
2. **Unoccupied, and the strongest single result.** The cavity-pulling read-out of Sec. 3: reading the microwave field as the frequency of a bad-cavity laser, so that cavity drift is rejected by κ/Γ instead of amplified. The underlying mechanism is standard active-optical-clock physics; what is new is applying it to electrometry, where drift amplification is the known weakness of every critical scheme.
3. **Unoccupied, but with near neighbours.** The null / threshold-crossing read-out. Biasing at a dynamical threshold and reading a frequency is not new in itself: Ref. [33] pulls a self-sustained microwave oscillator through its Adler synchronisation boundary, and Ref. [34] exploits a sub- to superradiant emission threshold for atomic state read-out. The novelty claim should be worded at the level of the specific realisation, not as “threshold-based sensing”.
4. **Unoccupied, and useful.** The observation of Sec. 5.1 that the incumbent NEPT is recovered from the lasing route by turning up the Rydberg interaction — i.e. that the two critical mechanisms are corners of a single phase diagram — and the gain-bandwidth characterisation of Sec. 4.

Item 2 is the one to lead with.

8 Assessment

8.1 Is it a good idea?

Yes, provided it is framed as a substitution rather than a scaling claim. The physics is sound, the map from measurand to phase boundary is exact and simple in both routes, the ingredients exist separately, and the numbers are competitive with — for Route I, better than — the incumbent.

What does *not* survive scrutiny is the framing “criticality gives super-classical scaling for electrometry”. Equations (24) and (12) show, for the two routes independently, that the floor is the ordinary $1/\sqrt{N}$ standard quantum limit; Sec. 4 shows that the $N^{4/3}$ ground-state advantage is removed by critical slowing down, in agreement with the known no-gos.

What does survive is sharper and more useful. Against the incumbent NEPT the light-matter routes offer (i) a critical point that is an *instrument setting* — repump and finesse, or laser power — rather than a property of the sample that drifts with temperature and density; (ii) an exact rather than calibrated scale factor; and, for Route I, (iii) active rejection of the dominant technical noise channel by $\kappa/\Gamma \approx 10^2$, which is the first answer anyone in this field has to the standing objection that criticality amplifies drift along with signal. Framed that way it is a good idea, and Route I is the one to build.

8.2 Benefits

- **Cavity-drift rejection (Route I).** $\kappa/\Gamma \approx 10^2$ suppression of the dominant technical channel, by the bad-cavity mechanism that underpins active optical clocks. This is the strongest single argument for the substitution.
- **A frequency read-out (Route I).** The measurand appears as an optical frequency at 1.124 MHz per (mV/cm), not as an amplitude — and all N atoms participate, giving a floor $5.1\times$ below Route II at equal N .
- **Transduction gain (Route II).** ε^{-1} gain, with roughly three decades of it free of bandwidth cost. Real Rydberg sensors are usually limited by optical detection and technical noise, not by atomic projection noise, so gain at the atomic stage is directly valuable.
- **In-situ tunable critical point.** Unlike the density-driven critical point of Refs. [6, 7], the boundary here is set by repump and finesse (Route I) or laser power (Route II) and can be servoed, making the operating point stable and reproducible.
- **SI traceability via Protocol B.** Reading the measurand as a laser detuning preserves the metrological virtue that makes Rydberg electrometry attractive to national metrology institutes.
- **Self-generated squeezing (Route II).** Up to 10 dB, with no external squeezed source and no separate entanglement-generation step. Route I, producing classical laser light, does not offer this.
- **One phase diagram, two corners.** The incumbent NEPT is recovered from Route I by turning up the Rydberg interaction (Sec. 5.1), so the proposal is a move within a known phase diagram rather than a competing claim.
- **Bright-to-dark operation.** The response sign is favourable for detection.
- **A trigger mode.** With strong Rydberg interactions the transition can be made first order, giving a latching threshold detector.

8.3 Risks and objections

- **The scaling advantage is illusory.** Established in Sec. 4.3.3 and by Refs. [19, 20, 22]. This is the objection a referee will raise first.
- **Criticality amplifies drift as well as signal.** Protocol A has no discrimination between $\delta\tilde{\omega}_0$ from the microwave field and $\delta\lambda$ from laser intensity noise or $\delta\omega$ from cavity drift — formally, the critical QFI matrix is sloppy [32]. Protocol C’s IF encoding and Protocol B’s null lock are the mitigations, and they are not optional.
- **Bandwidth.** 27.5 kHz at the parameters used, set by $(\kappa + \gamma)/2$. That is fine for metrology and useless for a communications receiver. A higher- κ cavity trades gain for bandwidth in the usual way.
- **Rydberg blockade caps N_{eff} ,** which is what Eq. (24) actually depends on. This is in direct tension with the Ref. [2] physics, which *wants* strong interactions.
- **Engineering overhead.** Two Raman channels plus a high-finesse cavity plus (probably) cold atoms is a substantial step away from the vapour-cell simplicity that makes Rydberg electrometry commercially interesting — though less of a step than it was, given Refs. [29, 31].
- **Decoherence cuts off the critical scaling.** The divergence is truncated once the soft-mode frequency falls below γ , i.e. at $\varepsilon_{\text{min}} \sim (\gamma/\omega)^2$; with vapour-cell linewidths this bites early.

- **Prior art is close.** Reference [7] in particular will be cited against any claim of novelty that is not carefully worded.

8.4 Recommended programme

1. **Build Route I.** It is the cheaper experiment, it reaches the lower floor, and its drift immunity is the defensible advantage. The apparatus already exists: Ref. [31] is a cavity-enhanced Rydberg superheterodyne receiver with a 19 dB gain, and what it lacks is a repump and a narrow transition. The first measurement to make is simply Eq. (6) — confirm that the emitted frequency tracks $\tilde{\omega}_0$ and ignores the cavity.
2. **Theory paper, correctly framed.** Something like “Microwave electrometry at a light-matter phase transition”. Lead with the substitution (Table 1), the drift-rejection result and the bridge back to the NEPT (Sec. 5.1); state the no-go results on critical scaling up front rather than being caught by them.
3. **Do the calculations a referee will ask for.** A Ziv–Zakai or Bayesian treatment of threshold-location precision, since the Cramér–Rao bound is the wrong tool for a null measurement; a proper account of inhomogeneous coupling across the cavity mode; and a full quantum (rather than semiclassical Maxwell–Bloch) treatment of Route I’s linewidth, since the mHz numbers in Sec. 3 rest on the standard bad-cavity Schawlow–Townes expression rather than on a calculation done here.
4. **Address the selectivity objection formally.** That criticality amplifies $\delta\lambda$ and $\delta\omega$ as readily as $\delta\tilde{\omega}_0$ is, in estimation language, the statement that the critical Fisher-information matrix is sloppy. Reference [32] shows for the Dicke model that multi-parameter critical estimation can retain divergent precision at the cost of a reduced exponent, and that a Dicke dimer with photon hopping restores the optimal scaling. Route I answers a large part of this objection physically rather than statistically, which is worth saying.
5. **Then Route II,** following the anisotropic-Dicke recipe of Ref. [29], for the ground-state critical point, the ε^{-1} gain and the squeezing.

A Numerical methods

All code is in `code/`; all figures are regenerated by `code/make_figs.py`.

Exact diagonalisation. Equation (13) is built on the product of the $j = N/2$ spin manifold ($N+1$ states) and a truncated Fock space. Since $\partial H/\partial\tilde{\omega}_0 = J_z$ is parity-even, only the even-parity sector contributes to F_Q and the calculation is restricted to it, which also removes the closing odd-parity gap from the eigensolver’s path. Photon truncation was verified at $N = 200$, $\lambda = \lambda_c$: F_Q agrees to twelve digits between $n_{\max} = 60$ and $n_{\max} = 120$; production runs used $n_{\max} = 100$ –140. The QFI is obtained from the fidelity susceptibility with a symmetric step $h = 10^{-4}\tilde{\omega}_0$.

Quantum Langevin. The linearised open-system calculation uses symmetrised two-sided spectra with vacuum inputs, diffusion matrix $D = \text{diag}(\kappa, \kappa, \gamma, \gamma)$, and the input–output relation $x_{\text{out}} = \sqrt{2\kappa}x_a - x_{\text{in}}$ including the cross-correlation between intracavity field and input vacuum. The implementation returns exactly $S_{\text{out}} = 1/2$ for an empty cavity, which fixes all normalisation conventions. The damped threshold λ_c is located by bisection on $\max \text{Re eig}(A) = 0$. Homodyne angles are optimised on a 361-point grid.

Parameters. Unless stated: $\omega/2\pi = \tilde{\omega}_0/2\pi = 1$ MHz, $\kappa/2\pi = 50$ kHz, $\gamma/2\pi = 5$ kHz, $d = 1774.82 ea_0$ for $^{87}\text{Rb } 53D_{5/2} \rightarrow 54P_{3/2}$.

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